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Suppose that $f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$. Similarly,

Express the right-hand side in term of z and $\bar{z}$. Using $x = \frac{z + \bar{z}}{2}$ and $y = \frac{z - \bar{z}}{2i}$.

Solution

$$x = \frac{z + \bar{z}}{2}, \quad y = \frac{z - \bar{z}}{2i}$$

$$\begin{aligned} f(z) &= x^2 - y^2 - 2y + i(2x - 2xy) \\ &= x^2 - y^2 - 2y + 2xi(1 - y) \\ &= \left(\frac{z + \bar{z}}{2}\right)^2 - \left(\frac{z - \bar{z}}{2i}\right)^2 - 2\left(\frac{z - \bar{z}}{2i}\right) \\ &\quad + 2i\left(\frac{z + \bar{z}}{2}\right)\left(1 - \frac{z - \bar{z}}{2i}\right) \\ &= \frac{z^2 + 2z\bar{z} + \bar{z}^2}{4} + \frac{z^2 - 2z\bar{z} + \bar{z}^2}{4} + i^2\left(\frac{z - \bar{z}}{i}\right) \\ &\quad + i(z + \bar{z})\left(1 - \frac{z - \bar{z}}{2i}\right) \\ &= \frac{z^2 + 2z\bar{z} + \bar{z}^2 + z^2 - 2z\bar{z} + \bar{z}^2}{4} + i(z - \bar{z}) \\ &\quad + i(z + \bar{z})\left(1 - \frac{z - \bar{z}}{2i}\right) \\ &= \frac{2z^2 + 2\bar{z}^2}{4} + iz - i\bar{z} \\ &\quad + \frac{1}{2}(2iz - z^2 + 2\bar{z} + 2i\bar{z} - z\bar{z} + \bar{z}^2) \\ &= \frac{z^2 + \bar{z}^2}{2} + iz - i\bar{z} + \frac{1}{2}(2iz + 2i\bar{z} - z^2 + \bar{z}^2) \\ &= \frac{z^2}{2} + \frac{\bar{z}^2}{2} + iz - i\bar{z} + iz + i\bar{z} - \frac{z^2}{2} + \frac{\bar{z}^2}{2} \\ f(z) &= \bar{z}^2 + 2i\bar{z} \end{aligned}$$

Suppose that $f(z_0) = g(z_0) = 0$ and that $f'(z_0)$ and $g'(z_0)$ exist, where $g'(z_0) \neq 0$.

Show that $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$.

Solution

$f(z_0) = 0$ and $g(z_0) = 0$ and $f'(z_0)$ and $g'(z_0)$ exist.

By the definition of derivatives,

$$\begin{aligned} f'(z_0) &= \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} \\ &= \lim_{z \rightarrow z_0} \frac{f(z)}{z - z_0} \end{aligned}$$

$$\begin{aligned} g'(z_0) &= \lim_{z \rightarrow z_0} \frac{g(z) - g(z_0)}{z - z_0} \\ &= \lim_{z \rightarrow z_0} \frac{g(z)}{z - z_0} \\ \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} &= \lim_{z \rightarrow z_0} \frac{\frac{f(z)}{z - z_0}}{\frac{g(z)}{z - z_0}} = \frac{\lim_{z \rightarrow z_0} \frac{f(z)}{z - z_0}}{\lim_{z \rightarrow z_0} \frac{g(z)}{z - z_0}} \\ &= \frac{f'(z_0)}{g'(z_0)} \end{aligned}$$

$$\therefore \lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}$$

3. Verify that the function

$f(z) = x^3 - 3x^2y + i(3x^2y - y^3)$ satisfies the Cauchy-Riemann equations and find its derivative

Solution

$$f(z) = x^3 - 3x^2y + i(3x^2y - y^3)$$

$$u(x, y) = x^3 - 3x^2y \quad \text{and} \quad v(x, y) = 3x^2y - y^3$$

$$u_x(x, y) = 3x^2 - 3y^2, \quad v_x(x, y) = 6xy$$

$$u_y(x, y) = -6xy, \quad v_y(x, y) = 3x^2 - 3y^2$$

$$u_x(x, y) = v_y(x, y) \quad \text{and} \quad u_y(x, y) = -v_x(x, y)$$

Therefore, $f(z)$ satisfies the Cauchy-Riemann equations.

$$\begin{aligned} f'(z) &= u_x(x, y) + i v_x(x, y) \\ &= 3x^2 - 3y^2 + i 6xy \\ &= 3(x^2 - y^2 + 2ixy) \\ &= 3(x + iy)^2 \\ &= 3z^2 \end{aligned}$$

4. Show that $f'(z)$ and its derivative $f''(z)$ exist everywhere and find $f'(z)$ and $f''(z)$ when $f(z) = \cos x \cosh y - i \sin x \sinh y$.

Solution

$$f(z) = \cos x \cosh y - i \sin x \sinh y$$

$$u(x, y) = \cos x \cosh y \quad \text{and} \quad v(x, y) = -\sin x \sinh y$$

$$u_x(x, y) = -\sin x \cosh y, \quad v_x(x, y) = -\cos x \sinh y$$

$$u_y(x, y) = \cos x \sinh y, \quad v_y(x, y) = -\sin x \cosh y$$

u_x, u_y, v_x, v_y are continuous everywhere in the complex plane.

$$u_x(x, y) = v_y(x, y), \quad u_y(x, y) = -v_x(x, y)$$

Therefore, the Cauchy-Riemann equations are satisfied for $f(z)$.

Thus, $f'(z)$ exists and $f'(z) = u_x + i v_x$

$$= -\sin x \cosh y - i \cos x \sinh y$$

$$\text{Let } F(z) = f'(z) = -\sin x \cosh y - i \cos x \sinh y$$

$$= U + i V$$

$$U(x, y) = -\sin x \cosh y, \quad V(x, y) = -\cos x \sinh y$$

$$U_x(x, y) = -\cos x \cosh y, \quad V_x(x, y) = \sin x \sinh y$$

$$U_y(x, y) = -\sin x \sinh y, \quad V_y(x, y) = -\cos x \cosh y$$

U_x, U_y, V_x and V_y are continuous everywhere in the complex plane.

$$U_x(x, y) = V_y(x, y), \quad U_y(x, y) = -V_x(x, y)$$

Therefore, the Cauchy-Riemann equations are satisfied for $F(z)$.

Thus, $F'(z)$ exists. That is $f''(z)$ exists.

$$f''(z) = F'(z) = U_x + i V_x$$

$$= -\cos x \cosh y + i \sin x \sinh y$$

$$= -(\cos x \cosh y - i \sin x \sinh y)$$

$$= -f(z).$$

show that $f'(z)$ and its derivative $f''(z)$ exist everywhere, find $f'(z)$ and $f''(z)$ when (i) $f(z) = iz + 2$,
(ii) $f(z) = z^3$.

Solution (i)

$$\begin{aligned} f(z) &= iz + 2 = i(x + iy) + 2 \\ &= ix - y + 2 \\ &= (2 - y) + ix \end{aligned}$$

$$u(x, y) = 2 - y, \quad v(x, y) = x$$

$$u_x(x, y) = 0, \quad v_x(x, y) = 1$$

$$u_y(x, y) = -1, \quad v_y(x, y) = 0$$

u_x, u_y, v_x and v_y are continuous everywhere in the complex plane.

$$u_x(x, y) = v_y(x, y) \text{ and } u_y(x, y) = -v_x(x, y).$$

Therefore, the Cauchy-Riemann equations are satisfied for $f(z)$.

$$\begin{aligned} \text{Thus, } f'(z) \text{ exists and } f'(z) &= u_x + iv_x \\ &= 0 + i1 \\ &= i \end{aligned}$$

$$\text{let } F(z) = f'(z) = i = 0 + i1$$

$$U(x, y) = 0, \quad V(x, y) = 1$$

$$U_x(x, y) = 0, \quad V_x(x, y) = 0$$

$$U_y(x, y) = 0, \quad V_y(x, y) = 0.$$

U_x, U_y, V_x and V_y are continuous everywhere in the complex plane.

$$U_x(x, y) = V_y(x, y) \text{ and } U_y(x, y) = -V_x(x, y).$$

Therefore, the Cauchy-Riemann equations are satisfied for $F(z)$.

Thus, $F'(z)$ exists.

That is $f''(z)$ exists.

$$\begin{aligned} f'(z) &= F'(z) = U_x + iV_x \\ &= 0 + i0 \\ &= 0 \end{aligned}$$

Solution (ii)

$$\begin{aligned} f(z) &= z^3 = (x + iy)^3 \\ &= x^3 + 3x^2iy + 3x(iy)^2 + (iy)^3 \\ &= x^3 + 3x^2iy - 3xy^2 - y^3 \\ &= (x^3 - 3xy^2) + i(3x^2y - y^3). \end{aligned}$$

$$u(x, y) = x^3 - 3xy^2, \quad v(x, y) = 3x^2y - y^3$$

$$u_x(x, y) = 3x^2 - 3y^2, \quad v_x(x, y) = 6xy$$

$$u_y(x, y) = -6xy, \quad v_y(x, y) = 3x^2 - 3y^2$$

u_x, u_y, v_x and v_y are continuous everywhere in the complex plane.

$$u_x(x, y) = v_y(x, y), \quad u_y(x, y) = -v_x(x, y).$$

The Cauchy-Riemann equations are satisfied for $f(z)$.

$$\begin{aligned} \text{Thus } f'(z) \text{ exists and } f'(z) &= u_x + iv_x \\ &= 3x^2 - 3y^2 + i6xy \\ &= 3(x^2 - y^2 + i2xy) \\ &= 3(x + iy)^2 \\ &= 3z^2 \end{aligned}$$

$$\begin{aligned} \text{let } F(z) &= f'(z) = 3z^2 = 3(x + iy)^2 \\ &= 3x^2 - 3y^2 + i6xy \end{aligned}$$

$$U(x, y) = 3x^2 - 3y^2, \quad V(x, y) = 6xy$$

$$U_x(x, y) = 6x, \quad V_x(x, y) = 6y$$

$$U_y(x, y) = -6y, \quad V_y(x, y) = 6x$$

U_x, U_y, V_x and V_y are continuous everywhere in the complex plane.

$$U_x(x,y) = V_y(x,y) \quad U_y(x,y) = -V_x(x,y)$$

Therefore, the Cauchy-Riemann equations are satisfied for $F(z)$.

Thus, $F'(z)$ exists. That is $F''(z)$ exists.

$$\begin{aligned} F''(z) &= F'(z) = U_x + iV_x \\ &= 6x + i6y \\ &= 6(x+iy) \\ &= 6z. \end{aligned}$$

6. Let the function $F(z) = \frac{1}{z}$. Prove that $F'(z)$ exists at any non-zero point $z = re^{i\theta}$.

Solution

$$F(z) = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r} e^{-i\theta} = \frac{1}{r} (\cos\theta - i\sin\theta)$$

$$u(r,\theta) = \frac{\cos\theta}{r}, \quad v(r,\theta) = -\frac{\sin\theta}{r}$$

$$u_r(r,\theta) = -\frac{1}{r^2} \cos\theta, \quad v_r(r,\theta) = \frac{1}{r^2} \sin\theta$$

$$u_\theta(r,\theta) = -\frac{\sin\theta}{r}, \quad v_\theta(r,\theta) = -\frac{\cos\theta}{r}$$

u, u_θ, v_r and v_θ are continuous everywhere.

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

The Cauchy-Riemann equations are satisfied.

$\therefore F'(z)$ exists.

$$\begin{aligned} F'(z) &= e^{-i\theta} (u + iv_r) \\ &= e^{-i\theta} \left(-\frac{1}{r^2} \cos\theta + i \frac{1}{r^2} \sin\theta \right) \\ &= e^{-i\theta} \left(-\frac{1}{r^2} \right) (\cos\theta - i\sin\theta) \\ &= e^{-i\theta} \left(-\frac{1}{r^2} \right) e^{-i\theta} \end{aligned}$$

$$F'(z) = -\left(\frac{1}{re^{i\theta}}\right)^2 = -\frac{1}{z^2} = -\frac{1}{z^2}$$

1. Show that the function $F(z) = \frac{1}{z^4}$ ($z \neq 0$) is differentiable in the indicated domain of definition and also to find $F'(z)$.

Solution

$$\begin{aligned} F(z) &= \frac{1}{z^4} \quad (z \neq 0) \\ &= \frac{1}{(re^{i\theta})^4} = \frac{1}{r^4 e^{i4\theta}} = \frac{e^{-i4\theta}}{r^4} \\ &= \frac{1}{r^4} (\cos 4\theta - i\sin 4\theta) \\ &= \frac{1}{r^4} \cos 4\theta - i \frac{1}{r^4} \sin 4\theta \end{aligned}$$

$$u(r,\theta) = \frac{1}{r^4} \cos 4\theta, \quad v(r,\theta) = -\frac{1}{r^4} \sin 4\theta$$

$$u_r(r,\theta) = -\frac{4}{r^5} \cos 4\theta, \quad v_r(r,\theta) = \frac{4}{r^5} \sin 4\theta$$

$$u_\theta(r,\theta) = -\frac{4}{r^4} \sin 4\theta, \quad v_\theta(r,\theta) = -\frac{4}{r^4} \cos 4\theta$$

u_r, u_θ, v_r and v_θ are continuous everywhere

$$ru_r = v_\theta \quad \text{and} \quad u_\theta = -rv_r$$

The Cauchy-Riemann equations are satisfied. $F'(z)$ exists.

$$\begin{aligned} F'(z) &= e^{-i\theta} (u_r + iv_r) \\ &= e^{-i\theta} \left(-\frac{4}{r^5} \cos 4\theta + i \frac{4}{r^5} \sin 4\theta \right) \\ &= -\frac{4}{r^5} e^{-i\theta} (\cos 4\theta - i\sin 4\theta) \\ &= -\frac{4}{r^5} e^{-i\theta} e^{-i4\theta} = -\frac{4}{r^5} e^{-i5\theta} \\ &= -\frac{4}{r^5 e^{i5\theta}} = -\frac{4}{(re^{i\theta})^5} = -\frac{4}{z^5} \end{aligned}$$

$z = re^{i\theta}$
 $ru_r = v_\theta, u_\theta = -rv_r$ C. R. Equations
 $F'(z) = e^{-i\theta} (u_r + iv_r)$ $u_x = v_y, u_y = -v_x$

Show that the function $f(z) = \sqrt{r} e^{i\theta/2}$ ($r > 0, \alpha < \theta < \alpha + 2\pi$) is differentiable in the domain of definition, and also to find $f'(z)$.

Solution

$$f(z) = \sqrt{r} e^{i\theta/2} \quad (r > 0, \alpha < \theta < \alpha + 2\pi)$$

$$= \sqrt{r} (\cos \theta/2 + i \sin \theta/2)$$

$$= \sqrt{r} \cos \theta/2 + i \sqrt{r} \sin \theta/2$$

$$u(r, \theta) = \sqrt{r} \cos \theta/2, \quad v(r, \theta) = \sqrt{r} \sin \theta/2$$

$$u_r(r, \theta) = \frac{1}{2} r^{-1/2} \cos \theta/2, \quad v_r(r, \theta) = \frac{1}{2} r^{-1/2} \sin \theta/2$$

$$u_\theta(r, \theta) = -\frac{1}{2} r^{1/2} \sin \theta/2, \quad v_\theta(r, \theta) = \frac{1}{2} r^{1/2} \cos \theta/2$$

u_r, u_θ, v_r and v_θ are continuous everywhere.

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

The Cauchy-Riemann equations are satisfied.

Thus $f'(z)$ exists.

$$f'(z) = e^{-i\theta} (u_r + iv_r)$$

$$= e^{-i\theta} \left(\frac{1}{2\sqrt{r}} \cos \theta/2 + i \frac{1}{2\sqrt{r}} \sin \theta/2 \right)$$

$$= \frac{1}{2\sqrt{r}} e^{-i\theta} (\cos \theta/2 + i \sin \theta/2)$$

$$= \frac{1}{2\sqrt{r}} e^{-i\theta} \cdot e^{i\theta/2}$$

$$= \frac{1}{2\sqrt{r}} e^{-i\theta/2}$$

$$= \frac{1}{2\sqrt{r} e^{i\theta/2}}$$

$$= \frac{1}{2f(z)}$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

9. Show that the function $f(z) = e^{-\theta} \cos(\ln r) + i e^{-\theta} \sin(\ln r)$, ($r > 0, 0 < \theta < 2\pi$) is differentiable in the indicated domain of definition, and also to find $f'(z)$.

Solution

$$f(z) = e^{-\theta} \cos(\ln r) + i e^{-\theta} \sin(\ln r)$$

$$(r > 0, 0 < \theta < 2\pi)$$

$$u(r, \theta) = e^{-\theta} \cos(\ln r), \quad v(r, \theta) = e^{-\theta} \sin(\ln r)$$

$$u_r(r, \theta) = -e^{-\theta} \sin(\ln r) \cdot \frac{1}{r}, \quad v_r(r, \theta) = e^{-\theta} \cos(\ln r) \cdot \frac{1}{r}$$

$$u_\theta(r, \theta) = -e^{-\theta} \cos(\ln r), \quad v_\theta(r, \theta) = -e^{-\theta} \sin(\ln r)$$

u_r, u_θ, v_r and v_θ are continuous everywhere.

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

The Cauchy-Riemann equations are satisfied.

Thus $f'(z)$ exists.

$$f'(z) = e^{-i\theta} (u_r + iv_r)$$

$$= e^{-i\theta} \left(-\frac{1}{r} e^{-\theta} \sin(\ln r) + i \frac{1}{r} e^{-\theta} \cos(\ln r) \right)$$

$$= e^{-i\theta} \left(-\frac{i}{r} e^{-\theta} \cos(\ln r) + i \frac{1}{r} e^{-\theta} \cos(\ln r) \right)$$

$$= \frac{i}{r e^{i\theta}} f(z)$$

$$= \frac{i}{z} f(z)$$

10. Prove that the function

$f(z) = 3x + y + i(3y - x)$ is entire.

Solution

$$f(z) = 3x + y + i(3y - x)$$

$$u(x, y) = 3x + y, \quad v(x, y) = 3y - x$$

$$u_x(x, y) = 3, \quad v_x(x, y) = -1$$

$$u_y(x, y) = 1, \quad v_y(x, y) = 3$$

$$u_x(x,y) = v_y(x,y) \quad u_y(x,y) = -v_x(x,y)$$

∴ The Cauchy - Riemann equations are satisfied.

∴ $f'(z)$ exists everywhere.

∴ $f(z)$ is analytic everywhere in the whole plane.

∴ $f(z)$ is entire function.

Ex 11 Prove that the function $f(z) = \sin x \cosh y + i \cos x \sinh y$ is entire and find $f'(z)$.

Solution

$$f(z) = \sin x \cosh y + i \cos x \sinh y$$

$$u(x,y) = \sin x \cosh y, \quad v(x,y) = \cos x \sinh y$$

$$u_x(x,y) = \cos x \cosh y, \quad v_x(x,y) = -\sin x \sinh y$$

$$u_y(x,y) = \sin x \sinh y, \quad v_y(x,y) = \cos x \cosh y$$

$$u_x(x,y) = v_y(x,y), \quad u_y(x,y) = -v_x(x,y)$$

∴ The Cauchy - Riemann equations are satisfied.

∴ $f'(z)$ exists everywhere.

∴ $f(z)$ is analytic everywhere in the whole plane.

∴ $f(z)$ is entire function.

$$f'(z) = u_x + i v_x$$

$$= \cos x \cosh y - i \sin x \sinh y$$

Ex 12 Prove that the function $f(z) = e^{iz}$ is an entire and $f'(z)$.

Solution

$$f(z) = e^{iz} = e^{i(x+iy)} = e^{ix} \cdot e^{-y}$$

$$= e^{-y} (\cos x + i \sin x)$$

$$= e^{-y} \cos x + i e^{-y} \sin x$$

$$u(x,y) = e^{-y} \cos x, \quad v(x,y) = e^{-y} \sin x$$

$$u_x(x,y) = -e^{-y} \sin x, \quad v_x(x,y) = e^{-y} \cos x$$

$$u_y(x,y) = -e^{-y} \cos x, \quad v_y(x,y) = -e^{-y} \sin x$$

$$u_x(x,y) = v_y(x,y), \quad u_y(x,y) = -v_x(x,y)$$

Thus, the Cauchy - Riemann equations are satisfied.

So, $f'(z)$ exists everywhere.

∴ $f(z)$ is analytic everywhere in the complex plane.

∴ $f(z)$ is an entire function.

$$f'(z) = u_x + i v_x$$

$$= -e^{-y} \sin x + i e^{-y} \cos x$$

$$= i e^{-y} (\cos x + i \sin x)$$

$$= i e^{-y} \cdot e^{ix}$$

$$= i e^{i(x+iy)}$$

$$= i e^{iz}$$

$$z = r e^{i\theta}$$

$$z = x + i y$$

exponential

Show that $\exp(2 \pm 3\pi i) = -e^2$.

Solution

$$\begin{aligned}\exp(2 + 3\pi i) &= e^{2+3\pi i} = e^2 \cdot e^{3\pi i} \\ &= e^2 (\cos 3\pi + i \sin 3\pi) \\ &= e^2 (-1 + i(0)) \\ &= -e^2 \\ \exp(2 - 3\pi i) &= e^{2-3\pi i} \\ &= e^2 \cdot e^{-3\pi i} \\ &= e^2 (\cos 3\pi - i \sin 3\pi) \\ &= e^2 (-1 - i(0)) \\ &= -e^2\end{aligned}$$

$\therefore \exp(2 \pm 3\pi i) = -e^2$

(14) $\exp\left(\frac{2+\pi i}{4}\right) = \sqrt{\frac{e}{2}} (1+i)$

show that
Solution

$$\begin{aligned}\exp\left(\frac{2+\pi i}{4}\right) &= \exp\left(\frac{1}{2} + \frac{\pi i}{4}\right) \\ &= e^{\frac{1}{2} + \frac{\pi i}{4}} = e^{\frac{1}{2}} \cdot e^{\frac{\pi i}{4}} \\ &= e^{\frac{1}{2}} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}) \\ &= e^{\frac{1}{2}} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}}\right) \\ &= \frac{\sqrt{e}}{\sqrt{2}} (1+i) \\ &= \sqrt{\frac{e}{2}} (1+i)\end{aligned}$$

$\therefore \exp\left(\frac{2+\pi i}{4}\right) = \sqrt{\frac{e}{2}} (1+i)$

15. Show that $\log(-1-\sqrt{3}i) = \log 2 - \frac{2\pi}{3}i$.

Solution

Let $z = -1-\sqrt{3}i$

Compare with $z = x+iy$

$x = -1 < 0, y = -\sqrt{3} < 0$

$$\begin{aligned}r = |z| &= \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{1+3} \\ &= \sqrt{4} = 2.\end{aligned}$$

$\theta = -(\pi - \tan^{-1}|\frac{-\sqrt{3}}{-1}|)$

$= -(\pi - \tan^{-1}(\sqrt{3}))$

$= -(\pi - \frac{\pi}{3}) = -\frac{2\pi}{3}$

$\log z = \log r + i\theta$

$\therefore \log(-1-\sqrt{3}i) = \log 2 - \frac{2\pi}{3}i$

16. Show that $\log i = (2n + \frac{1}{2})\pi i$,

$n = 0, \pm 1, \pm 2, \dots$

$\log 1 = 0$

Solution

Let $z = i$

Compare with $z = x+iy$

$x = 0, y = 1$

$i = 0 + (1)i$

$r = |z| = \sqrt{x^2 + y^2} = \sqrt{0^2 + 1^2} = 1$

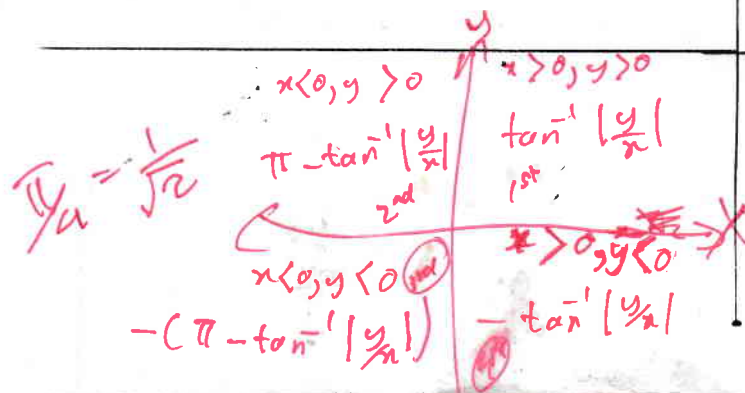
$\theta = \frac{\pi}{2}$

$\log z = \log r + i(\theta + 2n\pi), n = 0, \pm 1, \dots$

$\log i = \log 1 + i(\frac{\pi}{2} + 2n\pi)$

$= i(\frac{\pi}{2} + 2n\pi)$

$= (\frac{1}{2} + 2n)\pi i, n = 0, \pm 1, \pm 2, \dots$



$\log z = \log r + i\theta$

$\log z = \log r + i(\theta + 2n\pi), n = 0, \pm 1, \pm 2, \dots$

$r = |z| = |x+iy| = \sqrt{x^2 + y^2}$

$\theta = \tan^{-1}|\frac{y}{x}|$

17. Show that $\text{Log}(i^{1/2}) = (n + 1/4)\pi i$,
 $n = 0, \pm 1, \pm 2, \dots$

Solution

$$\begin{aligned}\text{Log}(i^{1/2}) &= \text{Log}(e^{\log i^{1/2}}) \\ &= \text{Log}(e^{1/2 \log i})\end{aligned}$$

Let $z = i$

Compare with $z = x + iy$

$x = 0, y = 1$

$r = |z| = \sqrt{x^2 + y^2} = \sqrt{0^2 + 1^2} = 1$

$\theta = \pi/2$

$\text{Log } z = \text{Log } r + i(\theta + 2n\pi), n = 0, \pm 1, \pm 2, \dots$

$\text{Log } i = \text{Log } 1 + i(\pi/2 + 2n\pi)$

$= (\pi/2 + 2n\pi)i, n = 0, \pm 1, \pm 2, \dots$

$\text{Log}(i^{1/2}) = \text{Log}(e^{1/2 (\pi/2 + 2n\pi)i})$

$= \text{Log}(e^{(\pi/4 + n\pi)i})$

$= i(\pi/4 + n\pi)$

$= i(\pi/4 + n\pi), n = 0, \pm 1, \pm 2, \dots$

$= (n + 1/4)\pi i$

18. Find the principal value of i^{-2i}

Solution

Principal value of $z^c = e^{c \log z}$

Principal value of $(i)^{-2i} = e^{-2i \log i}$

Let $z = i = 0 + i1$

Compare with $z = x + iy$

$x = 0, y = 1$

$\theta = \pi/2$

$r = |z| = \sqrt{0^2 + 1^2} = 1$

$\text{Log } z = \text{Log } r + i(\theta + 2n\pi), n = 0, \pm 1, \pm 2, \dots$

$\text{Log } i = \text{Log } 1 + i(\pi/2 + 2n\pi)$

Principal value of $(i)^{-2i}$

$= e^{-2i(\pi/2 + 2n\pi)i}$

$= e^{(\pi + 4n\pi)}$

$= e^{(1 + 4n)\pi}$

$= e^{(1 + 4n)\pi}, n = 0, \pm 1, \pm 2, \dots$

19. Find the principal value of $(-i)^i$

Solution

Principal value of $z^c = e^{c \log z}$

Principal value of $(-i)^i = e^{i \log(-i)}$

Let $z = -i = 0 + (-1)i$

Compare with $z = x + iy$

$x = 0, y = -1$

$r = |z| = \sqrt{0^2 + (-1)^2} = 1$

$\theta = -\pi/2$

$\text{Log } z = \text{Log } r + i\theta$

$\text{Log } (-i) = \text{Log } 1 + i(-\pi/2)$

$= -\pi/2 i$

Principal value of $(-i)^i = e^{i(-\pi/2 i)}$

$= e^{\pi/2}$

$= \exp \pi/2$

Principal value of $z^c = e^{c \log z}$

$e^{c \log z} = e^{c \log z}$

Show that $(1+i)^i = e^{(-\pi/4 + 2n\pi)} \cdot e^{(i \frac{\log 2}{2})}$

Solution

Principal value of $z^c = e^{c \log z}$

Principal value of $(1+i)^i = e^{i \log(1+i)}$

Let $z = 1+i$

Compare with $z = x+iy$

$$x=1, y=1$$

$$r = |z| = \sqrt{x^2+y^2} = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\theta = \tan^{-1} \left| \frac{y}{x} \right| = \tan^{-1} \left| \frac{1}{1} \right|$$

$$= \tan^{-1}(1)$$

$$\tan \theta = 1 = \tan \pi/4$$

$$\theta = \pi/4$$

$$\log z = \log r + i(\theta + 2n\pi), n=0, \pm 1, \pm 2, \dots$$

$$\log(1+i) = \log \sqrt{2} + i(\pi/4 + 2n\pi)$$

$$1+i = e^{\log \sqrt{2} + i(\pi/4 + 2n\pi)}$$

$$(1+i)^i = e^{i \log \sqrt{2} - (\pi/4 + 2n\pi)}$$

$$= e^{i \frac{1}{2} \log 2 - (\pi/4 + 2n\pi)}$$

$$= \exp(-(\pi/4 + 2n\pi)) \exp \frac{i \log 2}{2}$$

Since n takes all integral values, the term $-2n\pi$ here we can replace by $2n\pi$.

$$\therefore (1+i)^i = \exp(-\pi/4 + 2n\pi) \exp \left(\frac{i \log 2}{2} \right)$$

Find the principal value of i^i .

Solution

Principal value of $z^c = e^{c \log z}$

Principal value of $i^i = e^{i \log i}$

Let $z = i = 0 + i1$

Compare with $z = x+iy$

$$x=0, y=1, \theta = \pi/2$$

$$r = |z| = \sqrt{x^2+y^2} = \sqrt{0^2+1^2} = 1$$

$$\log z = \log r + i\theta$$

$$\log i = \log 1 + i(\pi/2)$$

$$= \pi/2 i$$

$$\text{Principal value of } i^i = e^{i(\pi/2 i)}$$

$$= e^{-\pi/2}$$

22. Find the principal value of

$$\left[\frac{e}{2} (-1 - \sqrt{3}i) \right]^{3\pi i}$$

Solution

Principal value of $z^c = e^{c \log z}$

Principal value of $\left[\frac{e}{2} (-1 - \sqrt{3}i) \right]^{3\pi i}$

$$= e^{3\pi i \log \left[\frac{e}{2} (-1 - \sqrt{3}i) \right]}$$

$$\text{Let } z = \frac{e}{2} (-1 - \sqrt{3}i) = -\frac{e}{2} - \frac{\sqrt{3}}{2} ei$$

Compare with $z = x+iy$

$$\therefore x = -\frac{e}{2}, y = -\frac{\sqrt{3}}{2} e$$

$$r = |z| = \sqrt{\left(-\frac{e}{2}\right)^2 + \left(-\frac{\sqrt{3}}{2}e\right)^2} = e \sqrt{\frac{1}{4} + \frac{3}{4}}$$

$$= e \sqrt{1} = e$$

$$\theta = -(\pi - \tan^{-1} \left| \frac{y}{x} \right|)$$

$$= -(\pi - \tan^{-1} \left| \frac{-\sqrt{3}e/2}{-e/2} \right|)$$

$$= -(\pi - \tan^{-1}(\sqrt{3})) = -(\pi - \pi/3)$$

$$= -2\pi/3$$

$$\log z = \log r + i\theta$$

$$\log \left[\frac{e}{2} (-1 - \sqrt{3}i) \right] = \log e + i(-2\pi/3)$$

$$= 1 - 2\pi/3 i$$

$$\text{Principal value of } \left[\frac{e}{2} (-1 - \sqrt{3}i) \right]^{3\pi i}$$

$$= e^{3\pi i (1 - 2\pi/3 i)}$$

$$= e^{3\pi i + 2\pi^2}$$

$$= e^{2\pi^2} \cdot e^{3\pi i}$$

$$= e^{2\pi^2} (\cos 3\pi + i \sin 3\pi)$$

$$= -e^{2\pi^2}$$

$$= -\exp(2\pi^2)$$

$$= \log 2^{1/2} - \pi/4 i$$

$$\log(1-i) = \frac{1}{2} \log 2 - \pi/4 i$$

$$\text{Principal value of } (1-i)^{4i}$$

$$= e^{4i (\frac{1}{2} \log 2 - \pi/4 i)}$$

$$= e^{2i \log 2 + \pi}$$

$$= e^{i 2 \log 2} \cdot e^{\pi}$$

$$= e^{\pi} [\cos 2 \log 2 + i \sin 2 \log 2]$$

23. Find the principal value of $(1-i)^{4i}$.

Solution

$$\text{Principal value of } z^c = e^{c \log z}$$

$$\text{Principal value of } (1-i)^{4i} = e^{4i \log(1-i)}$$

$$\text{Let } z = 1-i.$$

$$\text{Compare with } z = x+iy.$$

$$x=1, y=-1.$$

$$r = |z| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\theta = -\tan^{-1} \left| \frac{y}{x} \right| = -\tan^{-1} \left| \frac{-1}{1} \right|$$

$$= -\tan^{-1}(1)$$

$$\tan \theta = -1 = \tan(-\pi/4)$$

$$\theta = -\pi/4$$

$$\log z = \log r + i\theta$$

$$\log(1-i) = \log \sqrt{2} + i(-\pi/4)$$

- (24) Use the Cauchy-Riemann equations, to show that $f'(z)$ and its derivative $f''(z)$ exist everywhere and find $f'(z)$ and $f''(z)$ when $f(z) = e^{-x} \cdot e^{-iy}$.

Solution

$$f(z) = e^{-x} \cdot e^{-iy} = e^{-x} (\cos y - i \sin y) = e^{-x} \cos y - i e^{-x} \sin y$$

$$u(x, y) = -e^{-x} \cos y, \quad v(x, y) = -e^{-x} \sin y$$

$$u_x(x, y) = -e^{-x} \cos y, \quad v_x(x, y) = e^{-x} \sin y$$

$$u_y(x, y) = -e^{-x} \sin y, \quad v_y(x, y) = -e^{-x} \cos y$$

$$\therefore u_x(x, y) = v_y(x, y), \quad u_y(x, y) = -v_x(x, y)$$

$\therefore$ The Cauchy-Riemann equations are satisfied.

Thus, $f'(z)$ exists and $f'(z) = u_x + i v_x$.

$$f'(z) = -e^{-x} \cos y + i e^{-x} \sin y.$$

$$\text{Let } F(z) = f'(z) = -e^{-x} \cos y + i e^{-x} \sin y$$

$$U(x, y) = -e^{-x} \cos y, \quad V(x, y) = e^{-x} \sin y$$

$$U_x(x, y) = e^{-x} \cos y, \quad V_x(x, y) = -e^{-x} \sin y$$

$$U_y(x, y) = e^{-x} \sin y, \quad V_y(x, y) = e^{-x} \cos y$$

$$\therefore U_x(x, y) = V_y(x, y) \text{ and } U_y(x, y) = -V_x(x, y).$$

Therefore, the Cauchy-Riemann equations are satisfied for $F(z)$.

Thus, $F'(z)$ exists.

That is $f''(z)$ exists.

$$\begin{aligned} f''(z) &= F'(z) = U_x + i V_x \\ &= e^{-x} \cos y + i (-e^{-x} \sin y) \\ &= e^{-x} \cos y - i e^{-x} \sin y. \end{aligned}$$